# Extending design automation by integrating external services for product design

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*Abstract*—**Industry 4.0 aims at the promotion of design and manufacture automation to provide customized products. Knowledge-Based Engineering (KBE) is seen as a competitive way to realise design automation. This paper first introduces the Methodology and software tools Oriented to Knowledge-Based engineering Application (MOKA) method to develop an KBE application integrated with external calculators. Then a simple case of the wood-to-wood connection with different fasteners providing the connection capacity is demonstrated, showing the advantages to establish a model in parametric method integrated with external services. Meanwhile, this case shows the shortcomings of the integration of the current tools representing knowledge in different frameworks. Then a potential better way to integrate engineering knowledge is discussed.**

KBE; Knowledge integration; Parametric model; Design automation; Service-Oriented Architecture;

## I. Introduction

The customized products rather than standard products have become a trend. A vision of the Fourth Industrial Revolution (or Industry 4.0) is to meet the demand for mass production of customized products. Industry 4.0 is the digital transformation of production aiming at upgrading the traditional manufacturing to a high level of automation using modern smart technology [1]. Only with a more complete design and manufacture automation, the mass production of customized products can be realised.

Smart factory is a part of Industry 4.0, aiming at manufacturing automation through the implementation of the smart sensors, Internet of Things (IoT) and other related technologies [2]. Meanwhile, design automation tools are also an important part of the mass production of customized products, providing the feasible and manufacturable design from the customer's demand automatically. Currently, more and more design tools emerge to provide the customers with the freedom to design personalized products. This kind of design tools can be seen driven by Knowledge-Based Engineering (KBE) method. Based on the parametric templates of a product, the design tools give the users the access to change some of the parameters. However, the customized products at present are relatively simple, which means no complex engineering analysis is needed. When the customized products become more and more complex, the analyses in different domains need to be conducted to give the final result whether the design is feasible and manufacturable. Generally, it is difficult for a design software development team to understand and implement all the needed knowledge of different problem domains and integrate the analysis ability (knowledge) with the design tools. In order to provide the customers with a design tool which can generate a design solution for the following process automatically, the integration of external calculators (services) provided by other experts becomes a solution.

In this paper, the design of a wood-to-wood connection with different fasteners is chosen as an example to demonstrate the integration with the external calculators. Firstly, a practical way to develop a parametric model considering the external calculator is introduced. Secondly, the automatic design application which can generate the 3D model of wood-to-wood connection in Siemens NX Knowledge Fusion (the Knowledge-Based module) code is demonstrated to show how the external calculator is integrated. Finally, the benefit and limitation of the current integration is discussed and the advice on the future Knowledge-Based Engineering (KBE) application is given.

## II. State of the art

### A. KBE

KBE is the engineering using product and process knowledge that has been captured and stored in dedicated software applications, to enable its direct exploitation and re-use in the design of new products and variants [3]. One of the main features of KBE is the parametric model. Compared with the traditional CAD method, KBE method represents a model based on parameters and rules that can be embodied in a template. Knowledge organised in such form can be re-used relatively easily. Variants of an existing product can be designed rapidly, which makes the mass production of customized products possible.

With the significant benefit to support the rapid development of products, KBE support has been introduced into many CAD software products, for example, Autodesk Intent for Autodesk Inventor and Knowledge Fusion (KF) for Siemens NX. The example in this paper is represented in Knowledge Fusion code, which is an interpreted, object-oriented language that allows adding engineering knowledge to a part by creating rules which are the basic building blocks of the language

[4]. As an interpreted language, Knowledge Fusion is platform independent compared with compiled language, which is a helpful feature when deploying the KBE application in different platforms. KBE method has been implemented in some engineering products design. Tian et al. presented a new method based on KBE that can automatically generate the software components, which saves the designers from many repetitive (manual) processes [5]. Some researchers dedicated to extend the design ability of KBE applications. Saa et al. developed an ontology database integrated with experts knowledge and railway standards to support decision making for high-performance and cost-optimized design of complex railway portal frames [6]. Lobov et al. proposed the use of Knowledge Fusion for generation of robot trajectories to support faster transition from a product information in CAD and KF till the robot code that should weld the product [7]. In summary, KBE method has become a useful tool in design application development.

### B. Knowledge integration

Knowledge integration generally refers to the process of merging two or more originally unrelated knowledge structures into a single structure, which is the organic combination of the existing knowledge models of different sources [8] [9]. It is a generalized concept which appears in many fields, like database, machine learning, knowledge engineering and etc. For example, Shen et al. integrated railway signal maintenance knowledge from different sources to develop an application for maintenance knowledge push [9]. Here in this paper which is about a KBE application for product design, knowledge refers to the rules implemented in design process [10], so knowledge integration means to integrate the related knowledge from different domains used in design process into the automatic design application.

Service-Oriented Architecture (SOA) provides a good solution to integrate external service with the KBE design application. SOA is a software design method aiming to provide services to the other users through a communication protocol over a network, which is designed to be independent of platforms, products and technologies [11]. Such architecture makes each component of the application loosely coupled, which is easy to maintain and integrate new modules. This feature of SOA is helpful to dynamically meet the user's changing demands. When users propose some new demands, only the relevant new modules need to be added into the existing application without modifying other components. Some research proposed to implement SOA in KBE domain to provide customers the wanted service. Chen et al. proposed an ontology-based SOA framework to allow different legacy traditional Chinese medicine (TCM) information systems to interact, intercommunicate, and interoperate with each other [12]. Puttonen et al. proposes to implement semantic web services based on SOA to enable flexible manufacturing systems [13]. However, few research is about integrating SOA external calculators with design application based on KBE.

### C. Ontology-based engineering knowledge representation

One of the most important topics of intelligent systems is to represent the knowledge in the problem domain. The trend nowadays to represent engineering knowledge is to use ontologies and rules [6]. Ontology is designed to describe and identify the concepts to be shared in the semantic web. It enables reuse of domain knowledge and allow the users to modify the model without changing the database.

Some researchers implement ontology to represent engineering knowledge. Tran et al. proposed a method to implement ontology in KBE application development which is for 3D geometry generation, to gain the freedom to incorporate development tools regardless of discipline or platform [14]. Liu et al. presented an ontology-based framework to deal with the consistency in semantic representation of exchanged product knowledge in collaborative manufacturing [15]. Sanya et al. constructed the ontology model of their KBE system in the aerospace industry, which strengthens the knowledge reuse and eliminates platform-specific approaches to developing similar KBE systems [16]. These studies show that ontology-based engineering knowledge representation, as a higher level of abstraction, is helpful to deploy KBE application in different platforms.

### D. Summary

It can be seen that KBE method has been implemented in some engineering products design because the applications based on KBE method are easy to modify and integrate different knowledge domains. Meanwhile, SOA aims at providing services regardless of different platforms, which brings the convenience to integrate different design tools. This paper proposes an approach for creating KBE applications based on SOA that enables integration of heterogeneous services. Additionally, the potential benefits of implementing ontology-based engineering knowledge representation are discussed.

## III. METHODS

### A. Architecture

There are several KBE methodologies to develop the KBE applications, among which the Methodology and software tools Oriented to Knowledge-Based engineering Applications (MOKA methodology) is the most well-known one. This methodology divides the KBE application development into eight life-cycle steps, among which the two main focuses are "Capture" and "Formalize" [17] [18]. "Capture" is to collect and arrange the product knowledge in a structured form, which makes sure all the product parameters are obtained rightly. "Formalize" is to establish the model of product or process, in other words, to represent the knowledge using parameters and templates (rules).

The automatic design application in this paper is developed in the above-mentioned way. Aiming at easy maintenance and extension of more calculators, the Service-Oriented Architecture (SOA) is selected and illustrated in Fig. 1 and the sequence is illustrated in Fig. 2. The product knowledge is represented through data class and parametric templates stored in knowledge base server. The design control program which can be called ”designer” reads the data class to determine what to input, and then interacts with the user to receive the input parameters. After obtaining the needed input parameters, the designer program calls the external service to calculate the relevant performance and parses the returned result. Then the designer program passes all the parameters including geometric and non-geometric aspect to the DFA file generator. The generator passes the parameters into the parametric templates written in Knowledge Fusion code to generate the final model designed by the customer and store it in a file with DFA extension. Then the design of a customized product is completed in such way.

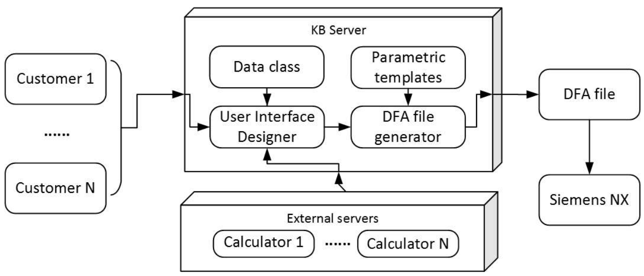


Fig. 1. The architecture of the automatic design application.

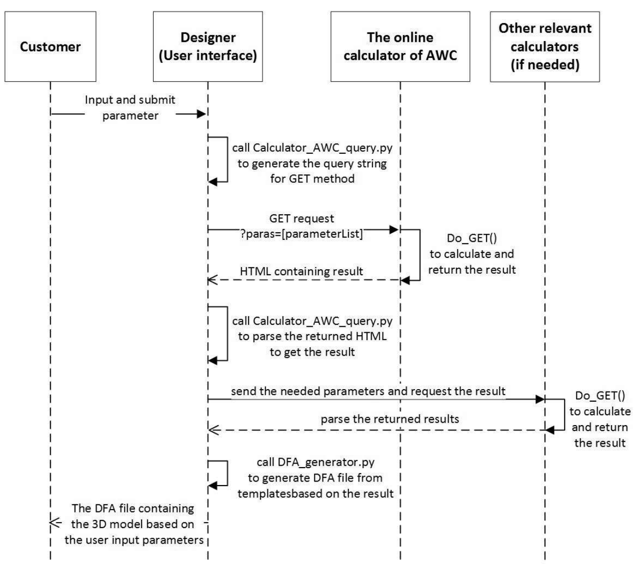


Fig. 2. The UML sequence diagram of the automatic design application.

### B. The parametric model integrated with external service

To realise the above-mentioned automatic design KBE application, the parametric model and the integration with the external service are the key points. This paper proposes to develop the automatic design application in following steps. Firstly, build the initial DFA file of the product to check if all the product parameters are captured. Secondly, parameterize the connection model and convert DFA draft file into parameters plus templates considering the parameters used in the online calculator. Thirdly, integrate the parametric model with the external service with suitable method, e.g. HTTP service. Then design the user interface to give the customer access to the templates based on the selected parameters. Finally, test and deploy the entire system for daily use.

## IV. Case study

### A. The case introduction

In order to demonstrate how the KBE application integrated with external calculator works in design automation, this paper proposes a simple case that two wood boards need to be connected in different ways providing sufficient lateral connection capacity. Wood-to-wood connection is widely used in furniture, which can be seen as a customized product based on the customer’s demand. Different connection ways including bolt, nail, lag screw, wood screw will be used for different purposes as each provides different lateral capacity and applies in different cases. (see Fig. 3)

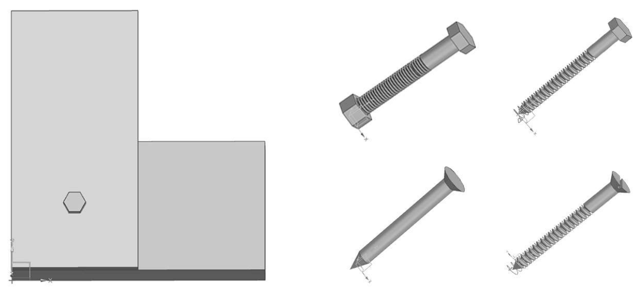

Fig. 3. Two wood boards and the four different fasteners.

Some professional organizations are providing online tools capable to calculate the connection capacity. American Wood Council (www.awc.org) provides users with a web-based approach to calculating capacities for single bolts, nails, lag screws and wood screws. Users select the connection parameters in the online user interface, e.g. the fastener type, the thickness of the main and side member, the diameter and length of different fasteners [19]. After user submission of the input parameters, the calculator will return the connection capacity, which is called the Adjusted ASD Capacity based on their professional algorithm. For simplification purpose, only part of the parameters concerned in the calculator will be considered in the model of this paper. (see Fig. 4)

For knowledge re-use purpose, an automatic design program which can provide the 3D model together with the connection capacity is developed using parametric modeling and online calculator. With this kind of tool, customers can design their wanted products by inputting the parameters of their own design and see whether they can be manufactured. Furthermore, if the automatic design program is connected with the smart factory, the customized products can be manufactured automatically after being ordered online, which is one of the visions of Industry 4.0.

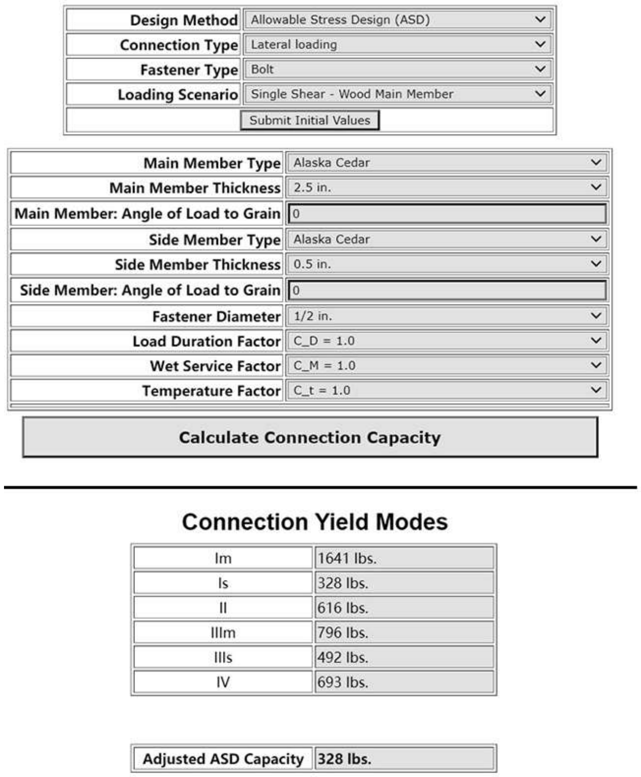


Fig. 4. The external online calculator to be integrated [19].

### B. Development of the automatic design program

The automatic design program is developed in the above-mentioned method. Firstly, the prototype of the wood-wood connection with different fastener types are modeled in Siemens NX Knowledge Fusion code, collecting and structuring all the needed engineering knowledge (geometric and non-geometric) to represent the product. The model code in this process can represent only one specific product and inconvenient to generate the variants. As the components used in the model are standard ones, if designer needs another fastener, he needs to input all the basic parameters by hand. To summarise, the parameter set are not simplified and the model are not well formalized. To formalize the model, a well-selected parameter set are needed. A good selection helps to reduce workload in the calculation part, otherwise a plenty of conversion will be needed. Thereby, the parameters are often selected considering the following process. If a parameter is always used as an input of the calculation, it is better to make it a basic parameter rather than a deduced one. A practical mean to select the parameters is to first select all the calculation needed parameters as basic and try to deduce the others from the basic ones. Then add those which are hard to deduce into the selection set (see table I).

In this paper, the deduced parameters are obtained through two means. Some can be calculated through formula, and some parameters that must be standard values are provided using a list storing the standard values in the template (see Fig. 5). After parameterizing the model, the external calculator can be integrated using HTTP service, which will be introduced later.

```
#! NX/KF 4.0
DefClass: twoboardtoconnect_woodScrew (ug_base_part);
(vector parameter) direction: vector(0,0,1);
(point parameter) start_point: point(45,0,0);
#the dimensions of the 2 boards
    (number parameter) L_1: <mm_length>;
    (number parameter) W_1: <mm_width>;
    (number parameter) Th_1: <mm_thickness_text>;
    (number parameter) L_2: <sm_length>;
    (number parameter) W_2: <sm_width>;
    (number parameter) Th_2: <sm_thickness_text>;
#the position to connect
    (point parameter) connect_point: point(min(L_1:, L_2:)/2,
       min(W_1:, W_2:)/2,Th_1: + Th_2:) + (start_point: - point(0,0,0));
#the required mechanical performance
    (number parameter) connection_capacity: <connection_capacity>;
    (number parameter) connection_requirement: <connection_requirement>;
    (number parameter) diameter_cylinder: <diameter_wood_screw>;
    (number parameter) length_wood_screw: <ls_length>;
#standard values stored in a list
    (list parameter)    head_thickness_list: {0.035, 0.043, 0.051,
                        0.059, 0.067, 0.075, 0.083, 0.091, 0.1, 0.108,
                        0.116, 0.132, 0.153, 0.164, 0.191, 0.196, 0.23};
    (list parameter)    diameter_list: {0.06, 0.073, 0.086, 0.099,
                        0.112, 0.125, 0.138, 0.151, 0.164, 0.177, 0.19,
                        0.216, 0.242, 0.268, 0.294, 0.32, 0.372};
```

Fig. 5. The excerpt of the DFA template.

TABLE I
PARAMETERS LIST.

| Parameter | Remark |
|---|---|
| main member thickness | inch |
| main member length | inch |
| main member width | inch |
| side member thickness | inch |
| side member length | inch |
| side member width | inch |
| fastener type | Bolt, Lag+Screw, Wood+Screw, Nail |
| bolt diameter | must be standard value |
| lag screw wash thickness | must be standard value |
| lag screw diameter | must be standard value |
| lag screw length | must be standard value |
| wood screw diameter | must be standard value |
| wood screw length | must be standard value |
| nail type | common wire, sinker, box |
| nail size | must be standard value |
| connection capacity | pound |
| connection capacity requirement | pound |

The auto-design program consists of 4 modules: user interface, data class, external service integrator, 3D model generator. Each of them is realised through a Python script:

- Designer.py: the design control program containing the user interface for parameters input.
- ConnectionClass.py: the data class containing the parameters (see table I).
- Calculator_AWC_query.py: the integrator that generates the query returning the result from the external service which is an online professional calculator, and also parses the returned result.
- DFA_generator.py: generate the 3D model (DFA file) using input parameters and result from the calculator.

The user interface (UI) module is used to guide the user to input the needed parameters which are defined in the data class and to call the other two modules to finish the design process.

For simplification purpose, the UI module in this paper is command-line style. However, it can be easily replaced by a graphical user interface (GUI) which can receive the user input in a more user-friendly way. As mentioned above, all the parameters are selected and defined in a data class, which is also beneficial for the future product upgrade, as the Object-Oriented template can be easily extended.

The external service integrator is the key to analyse the product performance. Given a professional online calculator, it is convenient for the user to get the exact connection capacity after inputting the design parameters properly. As shown in Fig. 2, this paper integrates the external calculator by implementing the "GET" method of HTTP protocol. The integrator script (Calculator_AWC_query.py) receives the user-input parameters transferred from the UI module and generates a query URL containing the parameters, then requests the data from the server by sending the generated URL. After receiving the request data containing the result, the integrator script parses it and returns the wanted connection capacity value to the designer script. In this paper, only one external calculator is integrated. However, in some complex design process, designers need to consider not only engineering performance, but also design standards, regulatory and safety codes, product cost and etc. [20]. Thus more calculators need to be integrated with the designer program.

After the user input and calculation process, all the needed data are prepared and transferred to the 3D model generator. This paper uses Siemens NX Knowledge Fusion to generate the product model (stored in a file with DFA extension that contains Knowledge Fusion code). The DFA generator fills in the parametric templates with the processed data by replacing the parameter placeholders, then generates the wanted DFA file containing the product model. User can visualise the model in Siemens NX for demonstration purpose.

### C. Result

The design tool can generate the 3D model of the connection with different fasteners automatically based on the user input. Fig. 6 shows the different models given by the design tool. When the model is in green, it means this design can provide a greater connection capacity than the user requirement (user input in advance). When in red, this design fails, indicating a better design is needed. This can be useful when customers or engineers design their customized products, which contributes to the future production of customized products.

## V. Discussion

The automatic design tool based on KBE method and integration with external service has a plenty of advantages. The tool in this paper can quickly generate a 3D model (DFA files) according to the customers' demands. Integrated with existing professional calculator, this tool can provide reliable calculation results. During the application development process, the software engineers are not required to understand the problem domain knowledge very well due to the knowledge re-use feature. In summary, this method gives a possibility for integrating a design tool with external services, promoting the customized product design.

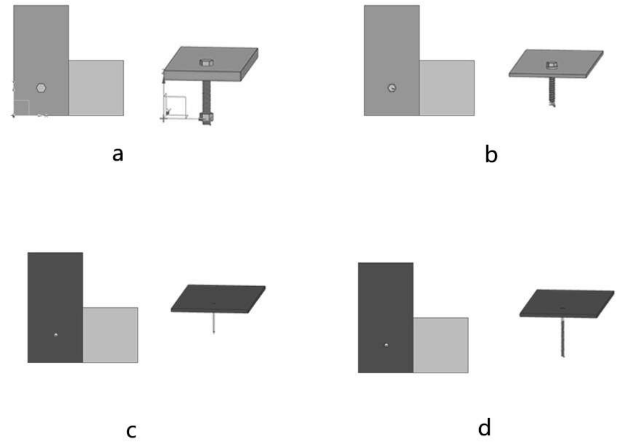

Fig. 6. Four connection designs with different fasteners: (a) bolt; (b) lag screw; (c) nail; (d) wood screw.

However, there are also some limitations of current framework.

- There are different representing rules in different modules. For instance, at first, the local design module and the external calculator use different parameter names and parameter set to represent the connection. In order to simplify the program, the rules in local design modules are modified to keep the same with the external calculator. However, this can be difficult when a lot of external services get involved.
- As the external calculator in this paper is not designed in a SOA way, it can be time-consuming and unrobust to extract the calculation result.
- Some reusable knowledge involved in this design are only human readable, needed to convert to different schema, e.g. the dimensions of the standard components (bolts and etc.) are stored in standard files rather than a digital schema.
- The parametric model is not dynamic, in other words, it can be hard for the user to add new features into the templates, which may restrict the design freedom.
- There are still some "if-then" rules in the templates, which may limit the handling of complex products.

Ontology-based machine-readable model is seen as a potential solution to the above-mentioned difficulties. It can represent product knowledge in a dynamic form, providing the convenience to the user to extend the templates. When the product becomes complex, it provides an efficient way to query the proper rules in the knowledge base compared with the "if-then" style. Furthermore, if ontology becomes a widely used way to represent knowledge, it can help to provide the exchangeable and machine-readable knowledge and SOA style calculators.

Overall, it is envisioned, that standardization or professional organizations, like in the case of using the online calculators from the American Wood Council [19], make it possible to dynamically integrate and extend product design with expert information provided by such organizations. As, for example, connection capacity is calculated in this paper by an online request to the AWC calculator from the CAD software getting back results that are immediately shown in the 3D model at the site of a product designer for the feasibility check.

## VI. Conclusions

In this paper, an architecture allowing integration of external services (e.g. that by AWC) is outlined and demonstrated to very early product design phase, where products parameters can be updated or selected depending on performance of envisioned product. It becomes easy to integrate such knowledge and it is important to keep it with professional or standardization organization motivating those to extend and expose these types of "calculators" as online services, which are possible to integrate with CAD with the help of APIs.

In the future work the extending of the services demonstrating multidisciplinary approaches will be explored, where the "calculators" or the services can come from different expert groups.

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